Proportional Control Design using Root Locus on the z-Plane

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1.0 Learning Outcomes

After studying this ACE-Lab learning guide, you should be able to demonstrate the Learning Outcomes listed in Table 1. On completion of the ACE-Lab learning guide, revisit Table 1 and confirm whether each Learning Outcome has been achieved.

Table 1: *Learning Outcomes from the Root Locus on the z-Plane Learning Guide*

ID	Learning Outcome	Achieved
1.	Explain the effect of the proportional control gain K_p on the closed-loop pole locations of a discrete-time system on the z-plane, and on the system response.	☐
2.	Apply the rules of root locus to sketch the root locus of a discrete-time closed-loop control system on the z-plane, including the circle rule for two poles and a zero.	☐
3.	Use the Jury stability test to find the gain at which the root locus leaves the unit circle, and the range of K_p that gives a stable system.	☐
4.	Use the root locus on the z-plane, with MATLAB, to select K_p to meet a performance requirement, and explain how sampling changes the result compared with continuous time.	☐

2.0 Introduction

In the Proportional Control Design using Root Locus Learning Guide, the root locus was used on the s-plane to show how the closed-loop poles move as K_p increases, and to select a gain that meets the system performance criteria (SPC). In the z-Plane Stability and Performance Learning Guide, the z-plane was introduced, with the unit circle as the boundary of stability, and the Jury test was used to find the stable range of K_p.

This Learning Guide brings these together. The starting point of the root locus is the open-loop transfer function of the system, which can be continuous-time or discrete-time; here it is the discrete-time transfer function $G(z)$ of the system controlled by a digital controller. An important question arises:

How do the closed-loop poles of a digital control system move as the gain is increased, and when do they leave the unit circle?

Why is the Root Locus on the z-Plane Important?

The main reasons why the root locus on the z-plane is important are:

i. it shows the stability of a digital control system over the full range of K_p, with the unit circle as the stability boundary;
ii. each point on the root locus corresponds to a damping ratio and natural frequency, so K_p can be selected directly in discrete time to meet the SPC;
iii. it shows that sampling changes the closed-loop behaviour: a system that is stable for all gains in continuous time can become unstable when controlled digitally;
iv. it is the basis for designing digital PID controllers, which are implemented on microcontrollers.

3.0 Effect of the Proportional Control Gain

Consider the closed-loop control system in Figure 1, with negative unity feedback and a proportional control gain K_p in the feedforward path:

$$G_{CL}(z) = \frac{K_p G(z)}{1 + K_p G(z) H(z)} = \frac{K_p \dfrac{N(z)}{D(z)}}{1 + K_p \dfrac{N(z)}{D(z)}} = \frac{K_p N(z)}{D(z) + K_p N(z)} \tag{1}$$

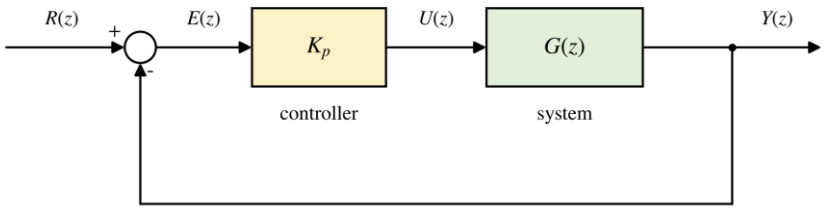


Figure 1: *Discrete-Time Closed-Loop Control System with a Proportional Control Gain K_p and Negative Unity Feedback*

Here $G(z)$ is the system $G(s) = 12/(s^2 + 5s + 6)$ discretised with a ZOH and a sampling interval of $T_s = 1$ s:

$$G(z) = \frac{N(z)}{D(z)} = \frac{1.3872z + 0.2561}{(z - 0.1353)(z - 0.0498)} \tag{2}$$

For three values of K_p, the closed-loop transfer functions are:

Table 2: *Closed-Loop Transfer Function and Poles for Three Values of K_p*

K_p	$G_{CL}(z)$	**Closed-loop poles**
0.1	$\frac{0.1387z + 0.0256}{z^2 - 0.0464z + 0.0324}$	$0.0232 \pm j0.1784$
0.4	$\frac{0.5549z + 0.1024}{z^2 + 0.3697z + 0.1092}$	$-0.1849 \pm j0.2739$
0.7	$\frac{0.9710z + 0.1793}{z^2 + 0.7859z + 0.1860}$	$-0.3929 \pm j0.1778$

Figure 2 shows that, as K_p increases, the poles move to the left on the z-plane, towards the negative real axis. The responses become faster, with a larger final value, but they become more oscillatory, and with $K_p = 0.7$ the output alternates from sample to sample. The final value is below 1 in each case, because the system is Type 0 and the controller is proportional only, so there is a steady-state error.

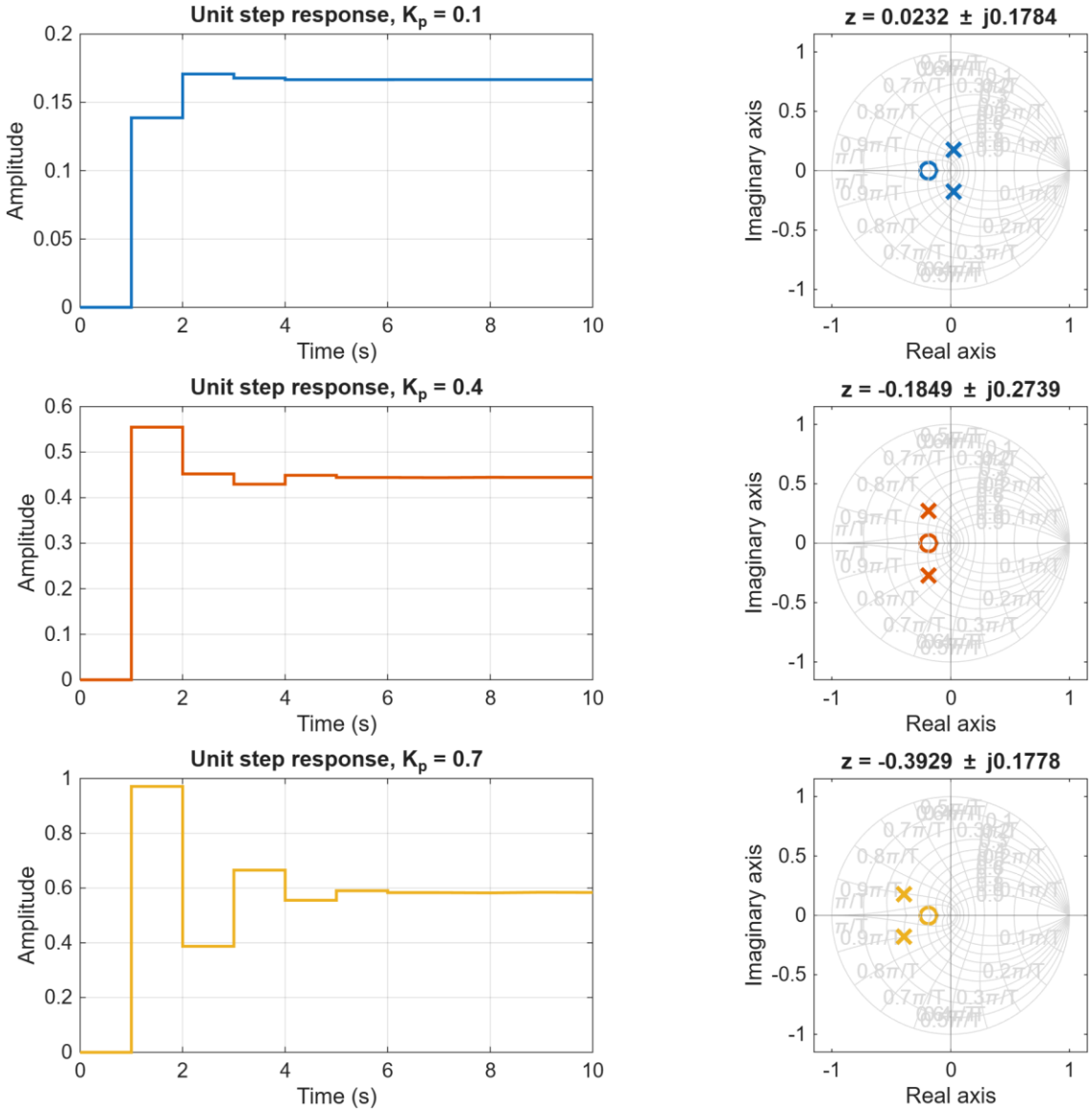


Figure 2: *Unit Step Responses and Closed-Loop Pole Locations for* $K_p = 0.1$, 0.4 *and* 0.7

4.0 Rules of Root Locus on the z-Plane

The rules of root locus assume negative unity feedback and a proportional control gain K_p in the feedforward path. The characteristic equation, $D(z) + K_p N(z) = 0$, has the same form as in continuous time, so the rules that build the root locus from the open-loop poles and zeros are unchanged. What changes is the interpretation: the stability boundary is now the **unit circle**, not the imaginary axis. The rules used here are listed in Table 3.

Table 3: *The Rules of Root Locus on the z-Plane*

Rule	Description
1	The root locus is symmetrical about the real axis.

Rule	Description
2	A segment of the real axis is part of the root locus if, at a given location along it, the total number of open-loop poles and zeros to the right is odd.
3	Each branch starts at an open-loop pole ($K_p = 0$) and ends at an open-loop zero or at infinity ($K_p = \infty$). The number of branches that end at infinity equals the relative degree, $n - m$, for a system with n poles and m zeros.
4	Where branches break away from the real axis, they leave at angles $\theta_{ba} = 360°/$(number of branches meeting) apart; for two branches, $180°$ apart, that is, perpendicular to the real axis.
5	For a second-order system with a zero to the left of two real poles, the complex part of the root locus is a **circle** centred on the zero, with a radius equal to the geometric mean of the distances d_1 and d_2 from the zero to each of the two poles: $r = \sqrt{d_1 d_2}$.
6	The Jury test is used to determine the value of K_p at which the closed-loop system becomes unstable, that is, where the root locus crosses the unit circle.
7	MATLAB is used for precise values: the gain, damping ratio and natural frequency at any point of the root locus.

The breakaway and break-in points of the circle in Rule 5 lie on the real axis at $z = z_0 \pm r$, where z_0 is the zero. For systems without a zero, or of higher order, the asymptote rules from the continuous-time Learning Guide still apply.

Example 1: Root locus of a second-order discrete-time system

For the open-loop transfer function of Equation (2), apply the rules of root locus, sketch the root locus on the z-plane, and determine the range of K_p for a stable closed-loop system.

Solution:

The poles are at $z = 0.1353$ and $z = 0.0498$, and the zero is at $z = -0.2561/1.3872 = -0.1846$.

Rules 1 and 2: the segment between the two poles (one pole to the right) and the segment to the left of the zero (three to the right) are on the root locus.

Rule 3: the relative degree is $2 - 1 = 1$, so one branch ends at the zero and one at $-\infty$ along the negative real axis.

Rules 4 and 5: the branches meet between the poles and break away perpendicular to the real axis, on a circle centred on the zero. The distances from the zero to the poles are $d_1 = 0.0498 + 0.1846 = 0.2344$ and $d_2 = 0.1353 + 0.1846 = 0.3199$, so:

$$r = \sqrt{d_1 d_2} = \sqrt{(0.2344)(0.3199)} = 0.2738$$

The branches break away at $z = -0.1846 + 0.2738 = 0.0892$ and rejoin the

real axis at $z = -0.1846 - 0.2738 = -0.4585$, where one branch moves right to the zero and the other moves left along the real axis.

Rule 6: the characteristic equation is $z^2 + (1.3872K_p - 0.1851)z + (0.0067 + 0.2561K_p) = 0$. The Jury conditions are:

$$|a_2| < 1 \Rightarrow K_p < 3.88, \qquad P(1) = 0.8216 + 1.6433K_p > 0, \qquad P(-1) = 1.1918 - 1.1311K_p > 0$$

so the range of K_p values for a stable system is $0 < K_p < 1.05$. At $K_p = 1.05$ the branch moving left reaches $z = -1$ on the unit circle, as shown in Figure 3.

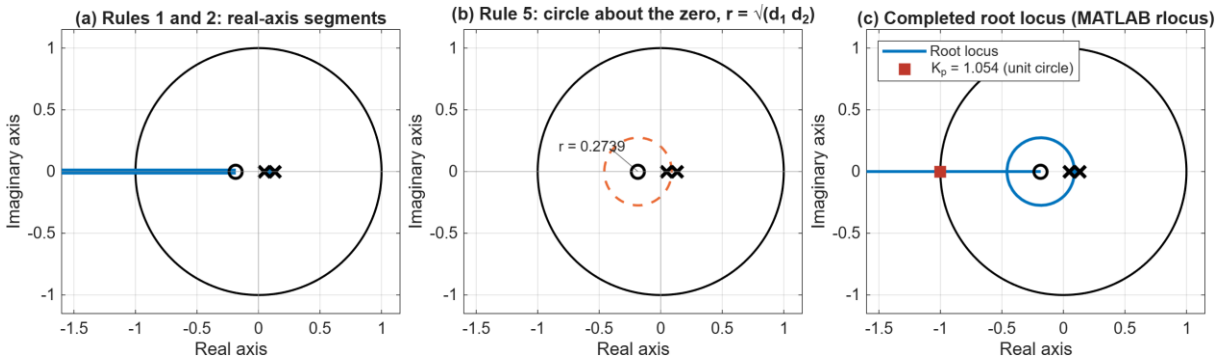


Figure 3: *Construction of the Root Locus for Example 1: (a) Real-Axis Segments, (b) the Circle about the Zero, (c) Completed Root Locus*

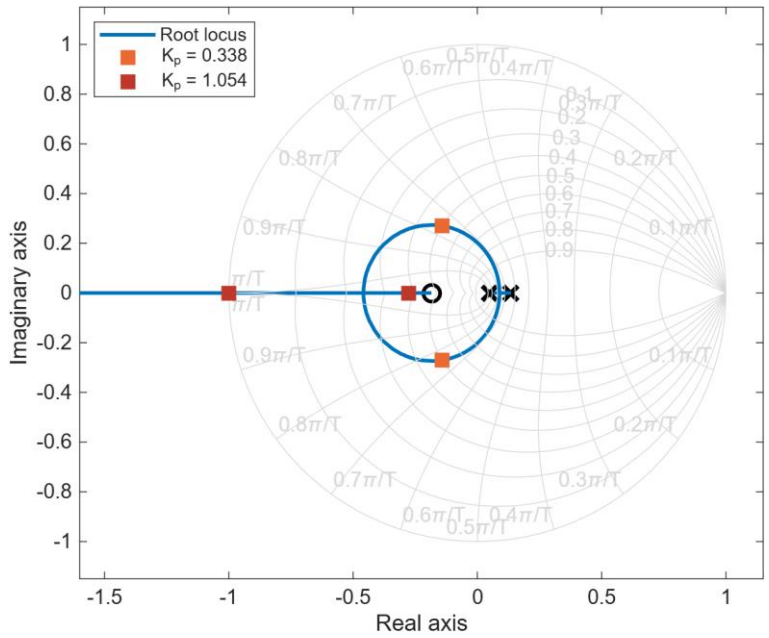
Using MATLAB: Root Locus and System Performance

MATLAB plots the root locus with the z-plane grid of constant damping ratio and natural frequency. Clicking on the root locus shows the gain, pole, damping and overshoot at that point. For example, a damping ratio of $\zeta = 0.5$ is reached at $K_p = 0.338$, with closed-loop poles at $z = -0.1419 \pm j0.2705$, and the stability limit is at $K_p = 1.05$, where the pole is at $z \approx -1$ with almost zero damping.

MATLAB Script 1: root locus on the z-plane for Example 1

```
clear; close all; clc;
Gs = tf(12,[1 5 6]);
Ts = 1;                     % sampling interval (s)
Gd = c2d(Gs,Ts,'zoh')       % discrete-time transfer function
rlocus(Gd)
zgrid
axis([-1.6 1.2 -1.2 1.2])
```

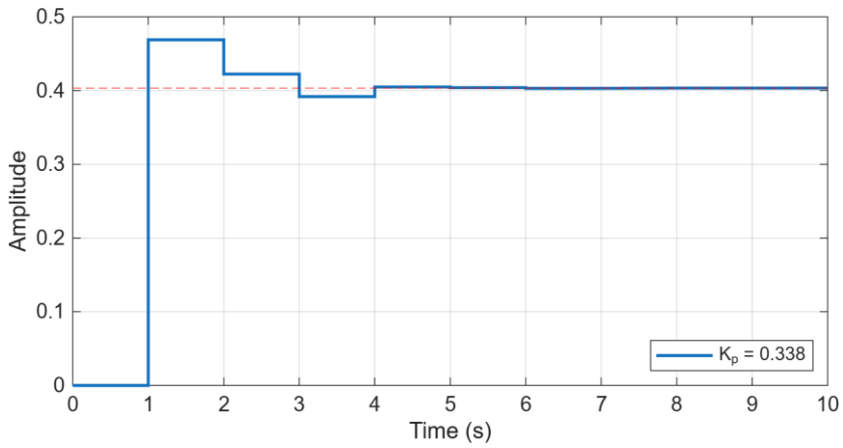
Output:



MATLAB Script 2: closed-loop step response with the selected gain

```
Kp = 0.338;
Gcl = feedback(Kp*Gd,1);
step(Gcl,10); grid on
stepinfo(Gcl)
```

Output:



The overshoot is 16% and the final value is 0.40. With a proportional controller only, the steady-state error, $1 - 0.40 = 0.60$, is large; integral action is needed to remove it, which is covered in the PID Learning Guides.

Example 2: DC motor angle control with a digital controller

The DC motor angle model from the continuous-time root locus Learning Guide, $G(s) = \frac{113.7}{s(s+3.398)}$, is controlled by a digital controller with $T_s = 0.05$ s. Sketch the root locus on the z-plane, determine the stable range of K_p, and

select K_p for a damping ratio of 0.7.

Solution:

Discretising with a ZOH gives:

$$G(z) = \frac{0.1344z + 0.1270}{(z-1)(z-0.8437)}$$

with poles at $z = 1$ (from the integrator) and $z = e^{-3.398(0.05)} = 0.8437$, and a zero at $z = -0.9450$.

Rules 1 to 5: the segment between the poles and the segment to the left of the zero are on the root locus. With two poles and a zero to their left, the complex part is a circle centred on the zero, with $d_1 = 1.7887$, $d_2 = 1.9450$ and:

$$r = \sqrt{(1.7887)(1.9450)} = 1.8652$$

The branches break away at $z = -0.9450 + 1.8652 = 0.9202$ and rejoin the real axis at $z = -2.8101$, far outside the unit circle.

Rule 6: the characteristic equation is $z^2 + (0.1344K_p - 1.8437)z + (0.8437 + 0.1270K_p) = 0$. The Jury conditions give:

$$|a_2| < 1 \Rightarrow K_p < \frac{1 - 0.8437}{0.1270} = 1.23, \qquad P(1) = 0.2614K_p > 0, \qquad P(-1) = 3.6875 - 0.0074K_p > 0$$

The digitally controlled motor is therefore stable only for $0 < K_p < 1.23$, although the continuous-time loop is stable for all K_p, because its relative degree is 2. The zero-order hold and sampling add an effective delay of about half a sampling interval, which bends the circle out through the unit circle.

Design: using MATLAB, the root locus crosses the $\zeta = 0.7$ line at $K_p = 0.0477$, with poles at $z = 0.9187 \pm j0.0765$. This is about 8% lower than the continuous-time design, $K_p = 0.0519$.

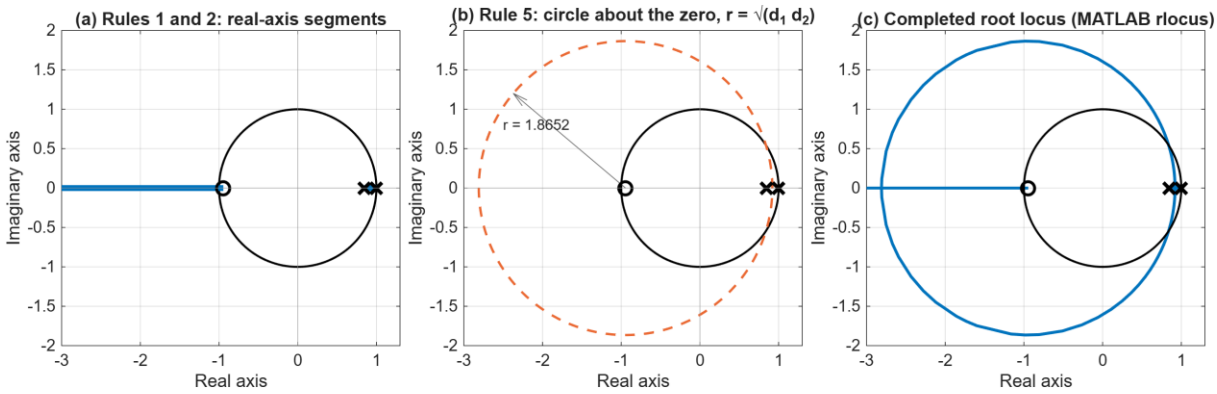


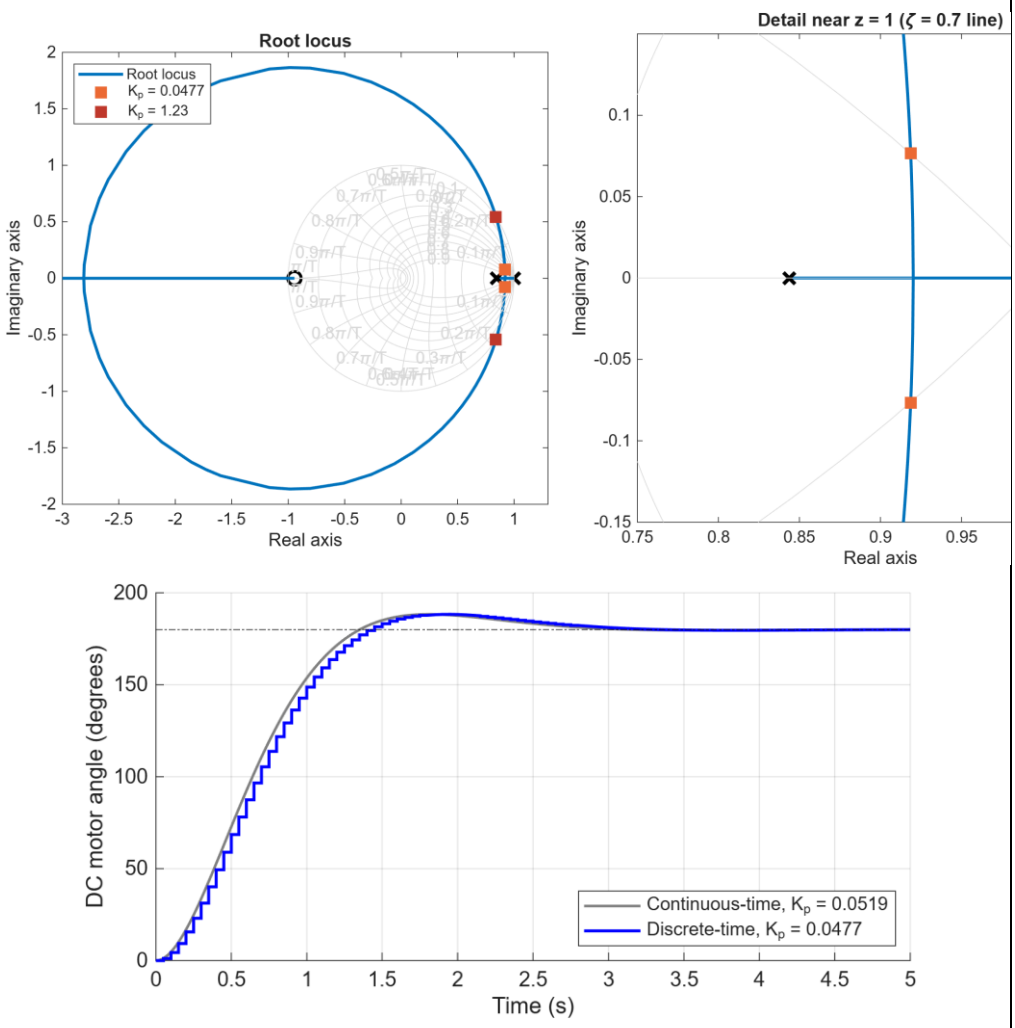
Figure 4: *Construction of the Root Locus for Example 2*

MATLAB Script 3: root locus and design for the digitally controlled DC

motor

```
clear; close all; clc;
Gs = tf(113.7,[1 3.398 0]);
Ts = 0.05;
Gd = c2d(Gs,Ts,'zoh')
rlocus(Gd); zgrid
%% closed-loop response with the selected gain
Kp = 0.0477;
Gcl = feedback(Kp*Gd,1);
opt = stepDataOptions('StepAmplitude',180);
figure; step(Gcl,opt,5); grid on
stepinfo(Gcl)
```

Output:



The sampled response has 4.6% overshoot and a rise time of 0.9 s, close to the continuous-time design.

Figure 5 shows the response with $K_p = 1.3$, just above the stability limit: the oscillation grows, although the same gain gives a stable, if oscillatory, response in continuous time.

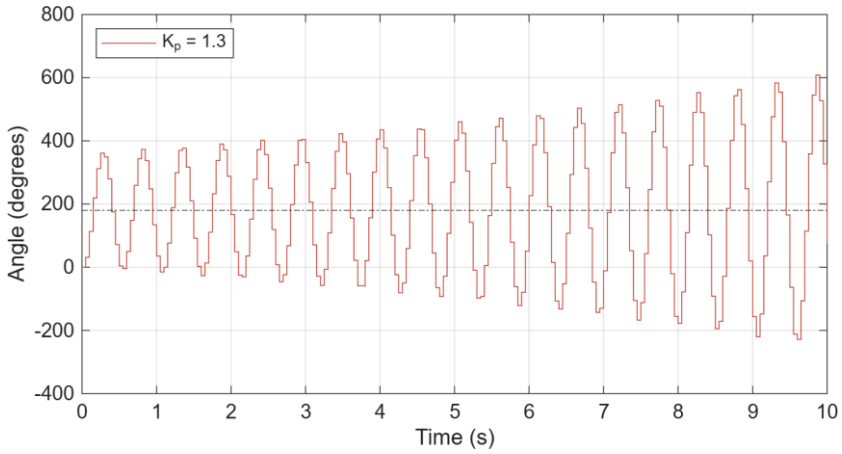


Figure 5: *Response of the Digitally Controlled DC Motor with $K_p = 1.3$, above the Stability Limit*

5.0 Exercises

For each of the following continuous-time systems, discretised with a ZOH at the given sampling interval and placed in a closed loop with negative unity feedback and a proportional control gain K_p: (i) determine $G(z)$ using MATLAB; (ii) sketch the root locus on the z-plane using the rules of root locus; (iii) determine the range of K_p for a stable system using the Jury test; and (iv) check the root locus using MATLAB.

a) $G(s) = \frac{2}{(2s+1)(s+4)}$, with $T_s = 0.1$ s.

b) $G(s) = \frac{12}{s^2+5s+6}$, with $T_s = 0.5$ s. Compare the stable range with that of Example 1, where $T_s = 1$ s.

c) $G(s) = \frac{5}{s+2}$, with $T_s = 0.2$ s.

d) $G(s) = \frac{1}{s(s+1)}$, with $T_s = 0.5$ s.

Fully worked solutions to the above exercises can be found in **Appendix A**.

6.0 Summary

This Learning Guide showed the effect of the proportional control gain K_p on a discrete-time closed-loop control system: on the pole locations on the z-plane, and on the system response. As K_p increases, the poles move away from their open-loop positions, and the response becomes faster but more oscillatory, until the poles leave the unit circle and the system becomes unstable.

The rules of root locus apply on the z-plane as on the s-plane, because the characteristic equation has the same form. For two real poles and a zero to their left, the root locus is a circle centred on the zero, with radius $\sqrt{d_1 d_2}$. The Jury test gives the gain at which the root locus crosses the unit circle, and MATLAB gives precise values for the design, allowing the system performance to be analysed for a given K_p.

Sampling matters. A digitally controlled system can become unstable at a gain that is safe in continuous time, and a shorter sampling interval brings the discrete-time design closer to the continuous-time one.

G(s) → G(z) → poles, zeros and real-axis segments → circle or asymptotes → Jury limit → select K_p on the z-plane grid

Appendix A: Fully Worked Solutions to Exercises

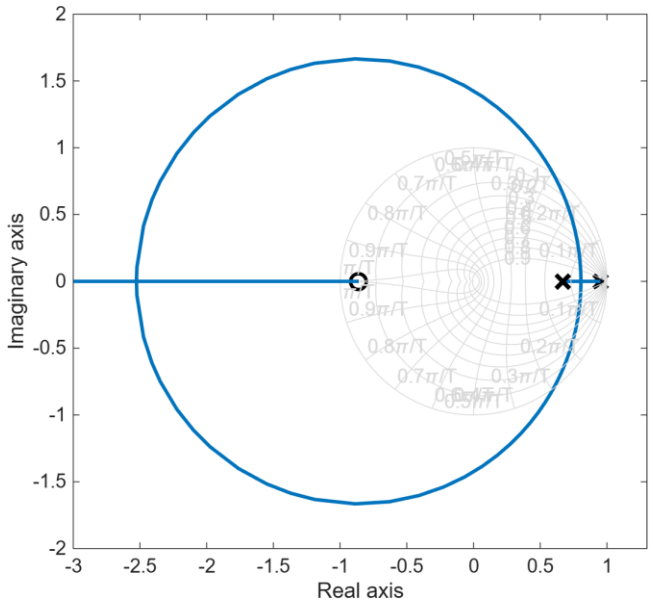
Question a:

```
Gd = c2d(tf(2,conv([2 1],[1 4])),0.1,'zoh')
```

$$G(z) = \frac{0.004321z + 0.003720}{(z - 0.9512)(z - 0.6703)}$$

Poles at 0.9512 and 0.6703; zero at -0.8608. The segments between the poles and to the left of the zero are on the root locus. The circle is centred on the zero with $r = \sqrt{(1.5311)(1.8120)} = 1.6657$, breaking away at 0.8049 and rejoining at -2.5265.

Characteristic equation: $z^2 + \left(0.004321K_p - 1.6215\right)z + \left(0.6376 + 0.003720K_p\right) = 0$. The binding Jury condition is $|a_2| < 1$, giving $0 < K_p < 97.4$; the poles then leave the unit circle at $z = 0.600 \pm j0.800$.



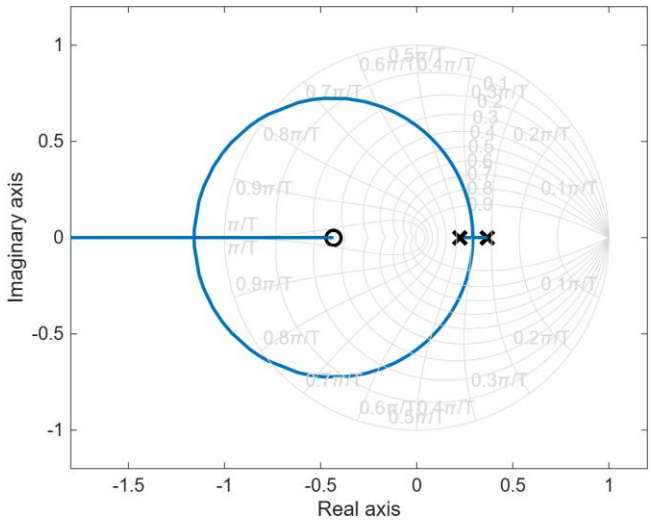
Question b:

$$G(z) = \frac{0.6852z + 0.2969}{(z - 0.3679)(z - 0.2231)}$$

Zero at -0.4333; circle radius $r = \sqrt{(0.6564)(0.8012)} = 0.7252$, breaking away at 0.2919 and rejoining at -1.1585.

Characteristic equation: $z^2 + (0.6852K_p - 0.5910)z + (0.0821 + 0.2969K_p) = 0$. Jury: $|a_2| < 1$ gives $K_p < 3.09$, and $P(-1) = 1.6731 - 0.3883K_p > 0$ gives $K_p < 4.31$. The system is stable for $0 < K_p < 3.09$, where the complex poles cross the unit circle at $z = -0.764 \pm j0.646$.

Halving the sampling interval from 1 s to 0.5 s increases the stable range from $0 < K_p < 1.05$ to $0 < K_p < 3.09$.



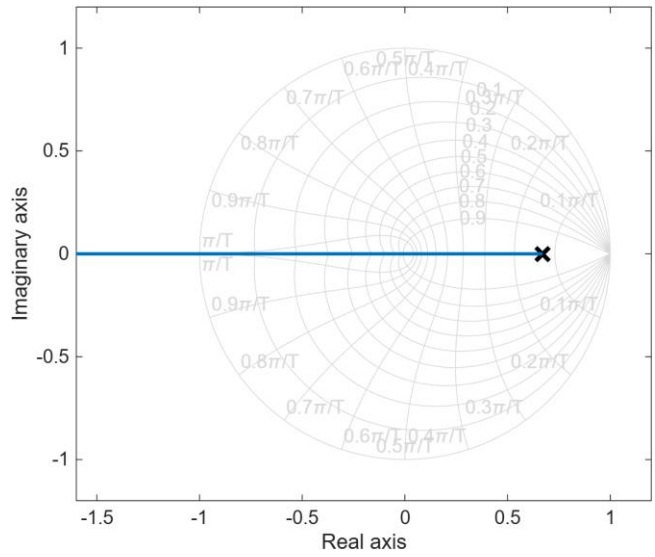
Question c:

$$G(z) = \frac{0.8242}{z - 0.6703}$$

There is one pole and no zero, so the single branch moves left along the real axis from 0.6703 towards $-\infty$. The closed-loop pole is at $z = 0.6703 - 0.8242K_p$, which must satisfy $|z| < 1$:

$$0.6703 - 0.8242K_p > -1 \;\Rightarrow\; K_p < \frac{1.6703}{0.8242} = 2.03$$

The system is stable for $0 < K_p < 2.03$. In continuous time, the first-order loop would be stable for all gains.

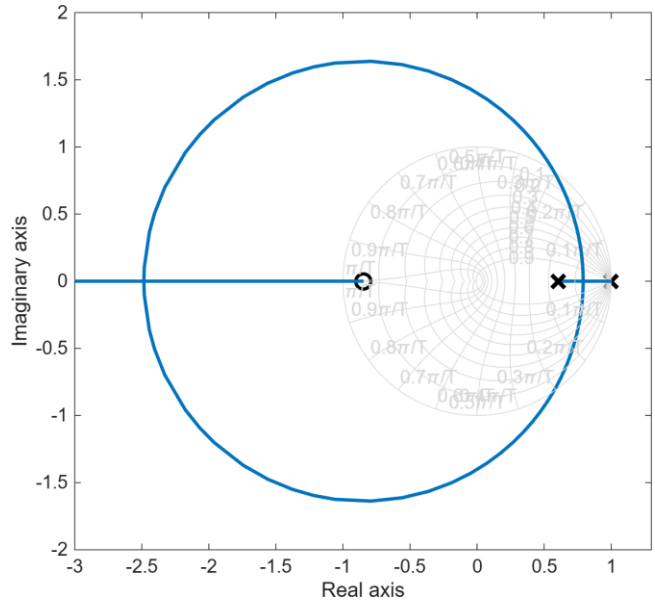


Question d:

$$G(z) = \frac{0.1065z + 0.0902}{(z-1)(z-0.6065)}$$

Poles at 1 and 0.6065; zero at -0.8467. Circle radius $r = \sqrt{(1.4532)(1.8467)} = 1.6382$, breaking away at 0.7915 and rejoining at -2.4850.

Characteristic equation: $z^2 + (0.1065K_p - 1.6065)z + (0.6065 + 0.0902K_p) = 0$. Jury: $|a_2| < 1$ gives $0 < K_p < 4.36$, with $P(1) = 0.1967K_p > 0$ and $P(-1) > 0$ for $K_p < 197$. The system is stable for $0 < K_p < 4.36$.



Recommended Citation

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